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A structural outline of the Yenkahe volcanic resurgent dome (Tanna Island, Vanuatu Arc, South Pacific)

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ABSTRACT

A structural study has been conducted on the resurgent Yenkahe dome (5 km long by 3 km wide) located in the heart of the Siwi caldera of Tanna Island (Vanuatu arc, south Pacific). This spectacular resurgent dome hosts a small caldera and a very active strombolian cinder cone – the Yasur volcano – in the west and exhibits an intriguing graben in its central part. Detailed mapping and structural observations make it possible to unravel the volcano-tectonic history of the dome. It is shown that, following the early formation of a resurgent dome in the west, a complex collapse (caldera plus graben) occurred and this was associated with the recent uplift of the eastern part of the present dome. Eastward migration of the underlying magma related to regional tectonics is proposed to explain this evolution.

1. Introduction

Resurgence is a common process in many calderas (Marsh, 1984; Newhall and Dzurisin, 1988). It corresponds to the uplift of the caldera floor (Smith and Bailey, 1968). The phenomenon has been recognized in active or recent calderas, such as Long valley, Valles caldera, Phlegrean Fields, Ischia, Iwo-Jima (e.g. Dvorak and Berrino, 1991; Orsi et al., 1996; Langbein, 2003; Ukawa et al., 2006; Feng and Newman, 2009; Vezzoli et al., 2009) as well as in old, eroded calderas (e.g. Seager, 1973; Beavon, 1980; Krupp, 1984; Fridrich et al., 1991; Kawakami et al., 2007; John et al., 2008). Detailed dating (Phillips et al., 2007; Kennedy et al., 2012) suggests that resurgence starts soon after caldera collapse.

Two main modes of resurgence have been identified: block uplift and doming (e.g. Acocella et al., 2001) In a resurgent block, the layers are uplifted more or less uniformly, they may be tilted but they are virtually undeformed, the deformation being accommodated by faults at the block borders. A well described example of block resurgence is that of Ischia (Orsi et al., 1991; Molin et al., 2003; Paoletti et al., 2013). For resurgent domes, most of the deformation is elastic with faults observed in the apex and peripheral areas. The well-known resurgent domes of Valles and Long Valley calderas (Bailey et al., 1976; Hill and Bailey, 1985; Heiken et al., 1990; Phillips et al., 2007) are examples of this type.

The mechanism of resurgence has been studied using different approaches: modelling of the deformation at active sites (e.g. Dvorak and Berrino, 1991; Luongo et al., 1991; Wicks et al., 1998; Orsi et al., 1999; Newman et al., 2006; Feng and Newman, 2009; Chang et al., 2010; Aly and Cochran, 2011), numerical modelling (Marsh, 1984; Chery et al., 1991), analogue modelling (Withjack and Scheiner, 1982; Marti et al., 1994; Acocella et al., 2001), and field observations of the inner parts of eroded systems (e.g. Smith and Bailey, 1968; Lipman, 1984; Fridrich et al., 1991; Kawakami et al., 2007). Several phenomena have been identified as potential triggers for resurgence: relaxation of the crust following caldera collapse (Chery et al., 1991), emplacement of intrusions (laccoliths, sills, plutons), and hydrothermal systems.

In presently active caldera resurgences, the vertical uplift ranges from a few centimetres per year (e.g. Long Valley) (Newman et al., 2006) to a few decimetres/meters per year (e.g. Phlegrean Fields, Iwo-Jima) (Ukawa et al., 2006; Troise et al., 2007). The study of ancient systems shows that the amplitude of resurgence may reach hundreds to thousands of metres. As a result, it has been proposed that the resurgent structures may become gravitationally unstable and could generate landslides (Tibaldi and Vezzoli, 2004). Caldera resurgence can develop in different modes, from continuous, long-term growth to discrete, shorter episodes of uplift. In general, active resurgences in calderas are associated with hydrothermal and seismic activity and periodic eruptions.

Thus the phenomena of caldera resurgence are important processes in volcanism and they involve interactions between magma, tectonics and hydrothermal systems. It has often been stressed that, in spite of their importance, the causes of presently active resurgences are usually not well understood. Because of their significance in terms of volcanic hazards, they constitute a major research issue. This is the main reason why we have launched a project on one of the most spectacular caldera resurgences in the world, that of the Yenkahe dome in the caldera of Siwi (Tanna Island, Vanuatu). Yasur volcano, a permanently active volcano, is located on the edge of the Yenkahe structure.

We present here observations and preliminary interpretations of the tectonic features of the Yenkahe. Our results suggest that the evolution

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of the resurgent structure is not a steady process, but that spatial displacements of the resurgence can be demonstrated. Their relationships with the active surface volcanism (collapse of a small caldera, construction of cinder cones, effusive activity) and internal processes (intrusions) are discussed.

2. Geological setting

The volcanism of the 1200 km long Vanuatu arc results from the subduction of the oceanic Indo-Australian plate beneath the oceanic Pacific plate to the east (Fig. 1A). The subduction rate is nearly 12 cm/yr at the latitude of Tanna Island located in the southern part of the arc (Pelletier et al., 1998; Calmant et al., 2003).

Tanna is a large construction, 60–80 km wide on the sea floor and about 2 km high (Robin et al., 1994) (Fig. 1B). The geology of Tanna Island (40×16 km) remains poorly known, in spite of a few pioneering works (Aubert de la Rüe, 1956; Carney and Macfarlane, 1979; Nairn et al., 1988; Robin et al., 1994; Chen et al., 1995; Neef et al., 2003; Allen, 2004; Peltier et al., 2012). In addition, only a few absolute ages are available to constrain the evolution of the island. The island is built up of volcanic products and coral reefs. Three volcanic complexes have been recognized by Carney and Macfarlane (1979): the Upper Pliocene to Pleistocene Green Hill to the north, the Pleistocene Tukosmeru volcano to the south, and the Siwi Group Volcanic Centre (Upper Pleistocene to present) to the east (Fig. 1, insert). In addition, Robin et al. (1994) have proposed the existence of a Pliocene–Pleistocene volcano, named Eastern Tanna volcano, whose centre would be located offshore, to the east of the northern part of the island.

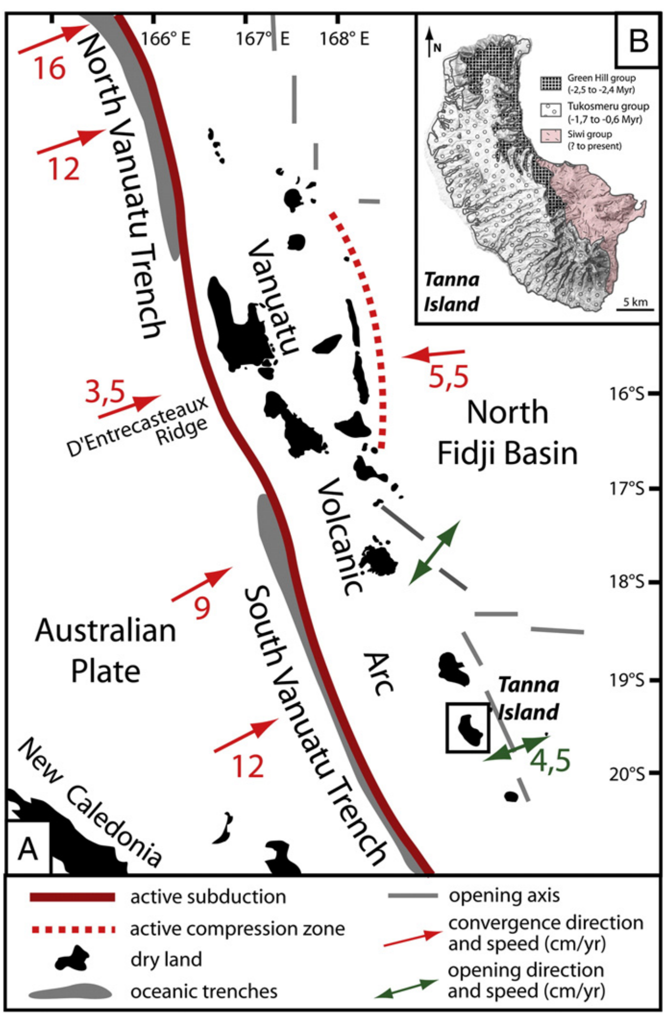

Fig. 1. A) The Vanuatu arc results from the subduction of the oceanic Indo-Australian plate beneath the oceanic Pacific plate to the east [after Pelletier et al. (1998) and Calmant et al. (2003)]. B) The volcanic island of Tanna belongs to the Vanuatu arc. Active volcanism and its emblematic Yasur volcano are located in the south-eastern part of the island [after Carney and Macfarlane (1979)].

3. The Siwi–Yasur–Yenkahe complex

The Siwi caldera is located at the base of a huge amphitheatre-like depression on the eastern flank of Tukosmeru volcano (Fig. 2). Morphologically, it appears as a relatively flat area, dipping gently to the east, bounded by faults a few tens of meters to nearly 100 m high. The caldera is open to the sea and therefore its eastern extent is unknown. The shape of the caldera is more or less rectangular (9×4 km) (Fig. 2). The formation of this caldera has been associated with the deposition of a sequence of pyroclastite deposits (Nairn et al., 1988; Robin et al., 1994; Allen, 2004). The caldera collapse is still undated, but the freshness of the deposits and the morphology suggest to some authors a relatively young age (less than 20 ky for Nairn et al., 1988).

The Yenkahe dome is a spectacular feature, ~5 km long by ~3 km wide and ~200 high, in the middle of Siwi caldera. It is elongated in the same direction as the caldera (Fig. 2). Although volcanic constructions exist on the Yenkahe, its main origin is attributed to resurgence processes. This is well illustrated by the presence of emerged corals and waterlain tuffs at its surface, up to 250 m above sea level (Carney and Macfarlane, 1979; Nairn et al., 1988; Chen et al., 1995; Neef et al., 2003). Chen et al. (1995) performed $^{230}Th/^{234}U$ dating on coral samples from terraces on the eastern part of the Yenkahe dome, at mean altitudes of 155 m and 15 m above sea level. Respective ages of AD 1002 $\pm$ 10 and AD 1868 $\pm$ 4 imply a mean uplift of 156 mm/yr for the eastern part of the Yenkahe dome in recent times. In addition, major deformation can be observed near its north-eastern and south-eastern borders where waterlain tuffs are tilted up to an angle of nearly 70° (Fig. 3D). The resurgence has been attributed to magma intrusion at depth (Carney and Macfarlane, 1979; Nairn et al., 1988; Chen et al., 1995; Metrich et al., 2011), and study of the emerged corals by Chen et al. (1995) suggests that the resurgence occurs mostly during discrete uplift events. This is well illustrated by the co-seismic uplift events of 6 and 4 m in 1878 (refs in Chen et al., 1995). The time averaged uplift rate of the dome over the last 1000 yr, as deduced from dating of the emerged corals (Chen et al., 1995), is 15.6 cm/yr. The Yenkahe resurgence is therefore one of the fastest of the presently active caldera resurgences (Geyer and Marti, 2010, and references therein). The morphology of the Yenkahe is not uniform and is described in detail in the following sections. Hydrothermally altered and warm ground spots are also frequent at the surface of the Yenkahe dome.

Two strombolian cones exist on the fringes of the Yenkahe dome. The Ombus cone (Fig. 2), located at the southern foot of the Yenkahe dome, is an extinct cone now heavily covered by vegetation. The Yasur cone is the present-day focus of magmatic activity in the Siwi Caldera (Figs. 2 and 3C), located at the north-western edge of the Yenkahe dome. It is 391 m above sea level, with a basal diameter of about 1000 m. The 100 – 150 m-deep crater at the summit is slightly elongated in a north-south direction and has an average diameter of 550 m. The three presently active vents at the bottom of the crater have frequent (every few minutes) strombolian explosions that expel ashes, scoriae, Pele's hair and centimetre-to-meter-sized bombs of basaltic–trachyandesitic composition (Metrich et al., 2011). The cone is built up of the accumulation of these products. To the north and west the deposits extend down to the Siwi caldera floor and the slopes of the cone are longer in these areas. Two historic landslides (1919 and 1975) occurred on these slopes (Carney and Macfarlane, 1979).

The Yasur's present cone is built on a former volcanic center from which the remnants of a pyroclastite cone and a caldera-like structure can be observed (Carney and Macfarlane, 1979). The remnants of the paleo-cone are located to the west-southwest of the present cone and

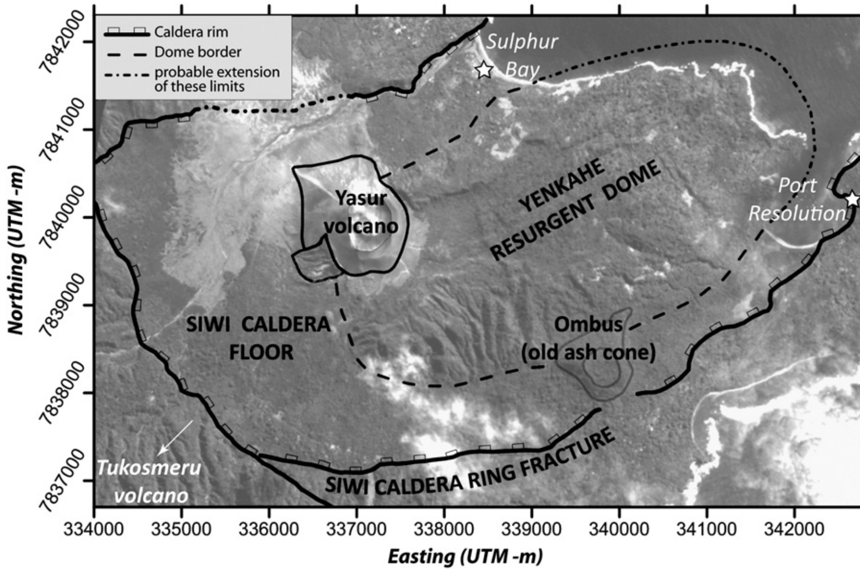

Fig. 2. The Yenkahe resurgent dome and the Yasur volcano are located within the Siwi Caldera.

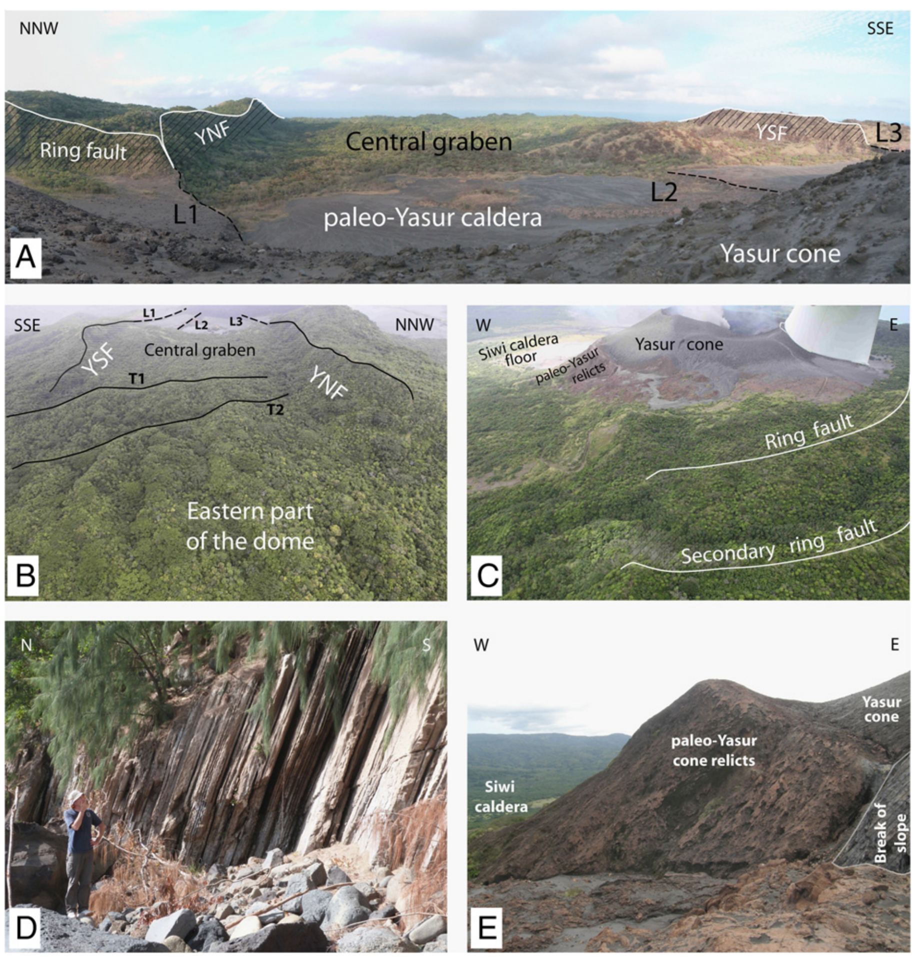

Fig. 3. A) The paleo-Yasur caldera and the central graben seen from the Yasur cone. The YNF and YSF delimit the central graben. The L1 and L3 faults within the paleo-Yasur caldera are exact extensions of the YNF and YSF but with different orientations. B) Aerial view of the central graben (photo courtesy of Adrien Normier and Clementine Bacri, ORA association). C) Aerial view showing the Yasur cone inside the paleo-Yasur caldera limited by the ring fault (photo courtesy of Adrien Normier and Clementine Bacri). D) Upturned marine formations along the north-eastern side of the Yenkahe dome in Sulphur Bay. E) Detail of the relict of the paleo-Yasur cone pasted onto the steeply dipping slope delimiting the paleo-Yasur caldera to the west (see photo C).

are progressively buried by the products of the recent cone (Fig. 3C and E).

Nairn et al. (1988) have proposed, on the base of the analysis and dating of two ash sequences to the northwest of Yasur, that the paleo-Yasur could have been active from 1400 to 800 yr ago, before the young Yasur started its activity.

4. Structural features of the Yenkahe dome

Three different structural domains can be identified on the Yenkahe dome:

- The western zone, which corresponds to the area occupied by the Yasur cone and remnants of the paleo-Yasur cone.
- The central zone, which is characterized by the presence of a well-defined axial graben
- The eastern zone, which is domed and extensively faulted

4.1. Western zone

The current boundary between the western and central zones is a semi-circular vertical fault (*Ring fault* on Fig. 4) (Fig. 3A and C). This fault, which cuts sharply and interrupts the central zone, is perpendicular to the main trend of the dome. It forms the eastern limit of the flat area in which the Yasur and paleo-Yasur cones have grown (Fig. 4).

Lava flows coming from the eruptive centres abut the fault, which clearly demonstrates that the fault predates lava infill in the western, flat area. Lava flows bounded by the fault probably correspond to the infilling of the depression by effusive activity of the paleo-Yasur subsequent to the fault movement. As noted by previous authors (Carney and Macfarlane, 1979), we consider this semi-circular vertical fault to be the eastern part of a ring fault, which surrounds a caldera-like structure. This caldera, located in the alignment of the axial graben, is nearly 4 km in diameter, which is noticeable. Yet, to our knowledge, no corresponding pyroclastite layers are associated with this caldera formation. We refer to this caldera as the paleo-Yasur caldera.

No ring fault is observed in the field along the western limit of the paleo-Yasur caldera. Instead, an abrupt change of the topographic surface, from flat to very steep, constitutes the end of the caldera floor (Fig. 4). This break in slope, which leads down to a hummocky-type topography in the floor of the Siwi caldera, may be interpreted as resulting from a landslide. It is important to note that the relict of the paleo-Yasur cone is pasted onto this steeply dipping slope (Fig. 4). This indicates that the steeply dipping slope that interrupts the paleo-Yasur caldera already existed prior to the formation of the paleo-Yasur cone.

However, these observations do not make it possible to determine whether landslides subsequently destroyed the western limit of the paleo-Yasur caldera or if the caldera was open to the west from its formation.

The northern and western limit of the paleo-Yasur caldera is hidden below the recent cone of the Yasur volcano. It is noteworthy that the Yasur cinder-cone is not circular in this sector and exhibits a strikingly straight line at the surface (Fig. 5). This geometric feature may result from the building of the Yasur cone onto pre-existing topography. Firstly, the northern slope of the Yasur cone mimics and extends the slope of the Yenkahe dome to the east. This suggests that the Yasur cone has been built on the preserved slope of the Yenkahe dome in this area. Secondly, the western slope of the Yasur cone straddles the western limit of the Paleo-Yasur caldera, i.e. the break in slope leading down to the floor of the Siwi caldera. Such pre-existing topography could explain the odd shape of the Yasur cone and the formation of the straight line in the cinder deposit at the junction of the northern and western slopes. This geometric feature (i.e. the straight line) is amplified by the building

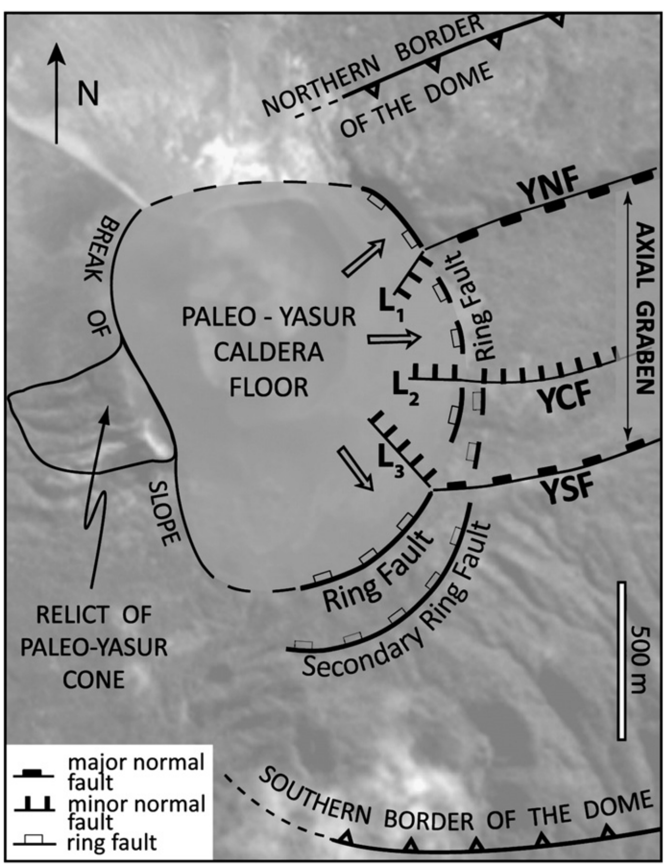

Fig. 4. The paleo-Yasur caldera is delimited to the east by a ring fault, which vanished in the west, being replaced by a break in slope leading down to the Siwi Caldera floor. A relict of the paleo-Yasur cone is pasted onto the break in slope at the western end of the paleo-Yasur caldera. Recent normal faults (L1, L2 and L3) converge towards the paleo-Yasur and the Yasur eruptive centres.

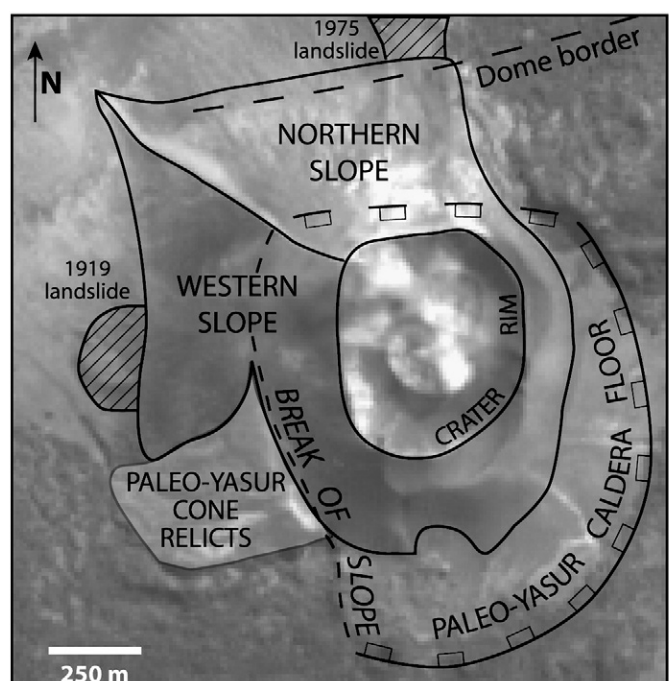

Fig. 5. The base of the Yasur cinder cone is roughly circular in the south-east but exhibits an odd extension to the northwest. Note the straight line dividing the north-western quadrant in two.

of a dune due to the dominant winds carrying the ashes to the northwest.

Lava infill within the paleo-Yasur caldera is cut by hydrothermally active normal faults (L1, L2 and L3 on Fig. 4) (Peltier et al., 2012), whose orientations converge towards the eruptive centre (Fig. 4). Strikingly, the two main faults (L1 and L3) observed in the lava infill are exact extension of the Yenkahe North Fault (YNF) and the Yenkahe South Fault (YSF), but with different orientations (Figs. 3A and 4).

To the south-east of the caldera ring fault, a secondary ring fault of limited offset can be identified, which follows exactly the same arc as the main ring fault (Figs. 3C and 4).

4.2. Central zone

The axial graben is limited to the north and south by two major normal faults (i.e. the YSF and the YNF, Figs. 3A,B and 6), both showing an offset, varying along strike from 40 to 60 m. Other normal faults of similar orientation and of a few-meters offset can be observed within the graben. One of them, however, the Yenkahe central fault (YCF), exhibits an offset of a few tens of meters to the west of the axial graben (Fig. 6).

The YNF is close to the northern border of the dome, which is narrow and exhibits a steep topographic slope (about 45°). In contrast, the YSF is far from the southern border and another striking normal fault can be observed to the south of it, in the generally flat area before the slope of the southern border. As a result, a simplified cross-section of the dome from north-west to south-east, including the central graben reveals an asymmetric structure (Fig. 7A).

Erosion and vegetation are very abundant in the axial central graben making the faults difficult to establish with certainty. The YNF is interrupted to the east by an oblique structure, which is interpreted as an erosional feature resulting from repeated landslides. The YSF can be traced eastward to a clear (offset to the order of a few tens of meters) curved transverse fault (T3 in Fig. 6), which marks the transition to another domain (see eastern zone section).

When approaching the paleo-Yasur caldera, the graben exhibits a perfectly flat floor resulting from the collapse of the axial zone by about 60 m. The flat floor of the graben is at exactly the same elevation than that of the paleo-Yasur caldera and no escarpment is visible in this zone when crossing the ring fault of the caldera (Fig. 3A).

To the east, the flat floor of the graben is interrupted by two transverse faults (Fig. 6, T1 and T2), which raise the topography locally. Due to the dense vegetation, the chronology between transverse and longitudinal faults cannot be established in the field. However, satellite images favour the hypothesis that transverse faults postdate longitudinal faults in the axial central graben.

4.3. Eastern zone

The altitude decreases gradually eastwards along the flat floor of the axial central graben. However, the graben floor is cut by a succession of transverse faults (faults T1, T2, T3, etc on Fig. 6), which all reveal a downthrow to the west and/or an upthrow to the east. This indicates a downward movement of the western part relative to the eastern part. In other words, even if the current topography shows an overall downwards slope to the east, the roof of the dome is uplifted to the east whereas it collapses to the west. As a result, the YNF and the YSF are less visible in the east, exhibiting small offsets, and the whole topography in the east is that of a classic resurgent dome exhibiting steep flanks (30 – 40°) with a flat roof in between (Fig. 7B).

Two sets of longitudinal and transverse faults are visible in the eastern part of the dome (Fig. 8). Longitudinal faults are oriented in the axial direction of the dome whereas transverse faults are associated with the roof uplift. No clear-cut chronology can be established either in the field or from satellite images for the two sets of faults, which may be considered as an argument for coeval activity.

The up-tilting of strata along the border of the dome, especially at Sulphur Bay (up to 70°, Fig. 3D), is consistent with very recent marine formations observed on top of the dome at an elevation of about 200 m. It is noteworthy that thermal anomalies are higher in the eastern zone than in the western part of the dome, especially along the dome borders where several hydrothermal zones are visible. These hydrothermally active zones are associated with local gravity landslides of highly altered rocks, as seen on the northern flanks of the dome at Sulfur bay

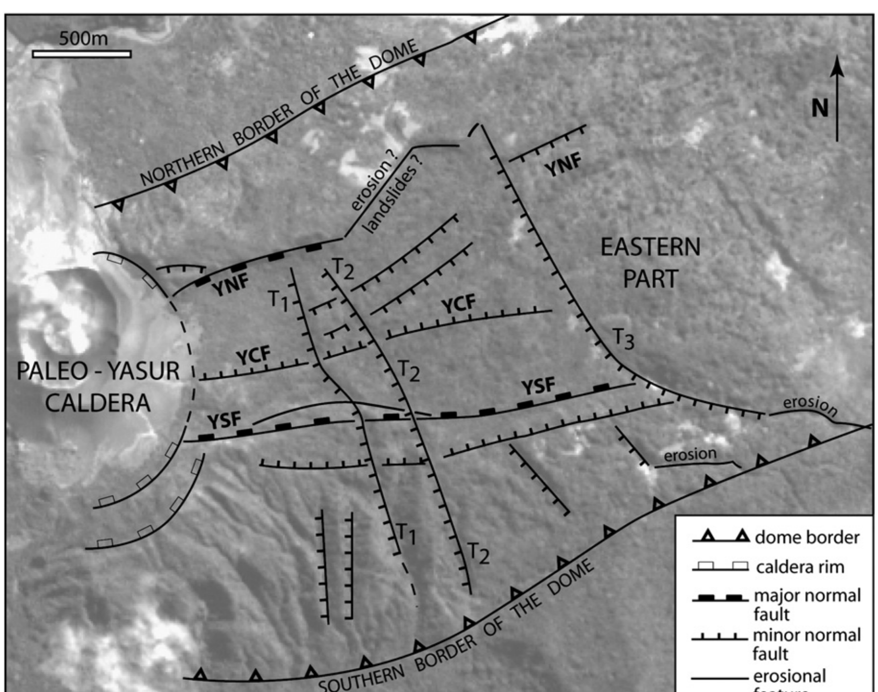

Fig. 6. The axial central graben is delimited by two major normal faults (YNF and YSF). Transverse faults (T1, T2 and T3) reveal an upthrow to the east associated with the late doming of the eastern part the Yenkahe dome.

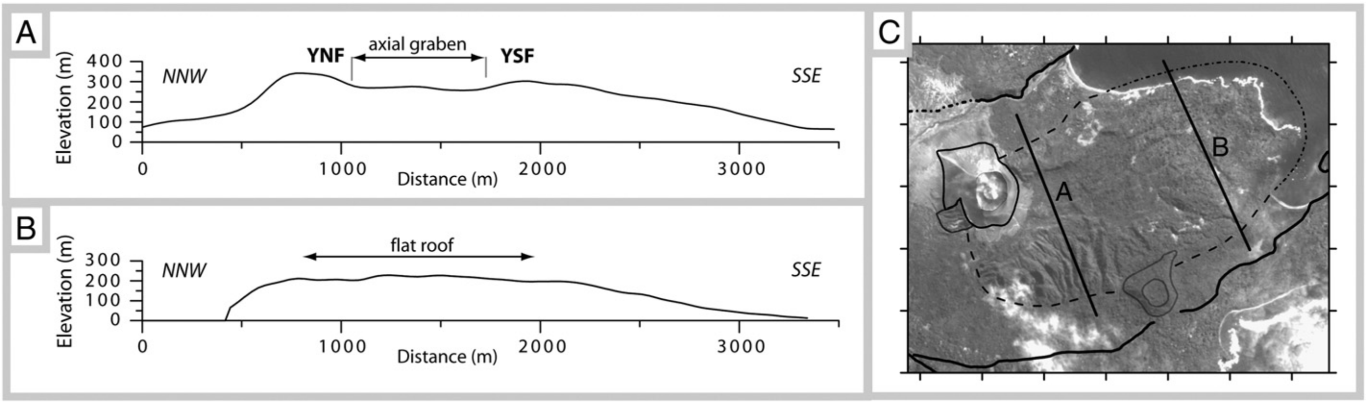

Fig. 7. Two cross-sections of the dome (see localisation in C) reveal the strong asymmetry of the Yenkahe dome with steep (40°) and gentle slopes (10°) to the north and south, respectively. A) To the west, the central part of the dome is collapsed between two main normal faults (YNF and YSF). B) In the east, the top of the dome is a flat zone bearing very recent marine deposits.

(Brothelande, PhD, in preparation). This suggests that the eastern part of the dome is active.

5. Discussion

5.1. Westward extension of the Yenkahe dome

The main structural observation in this area is that the dome is cut off by the ring fault of the paleo-Yasur caldera. This demonstrates that the formation of the dome occurred prior to that of the caldera and reveals that the western part of the Yenkahe dome collapsed during caldera formation.

The western extent of the dome is not clear. However, the abrupt break of slope to the west of the flat paleo-Yasur caldera floor is a clear limit beyond which, in the floor of the Siwi caldera, no relief corresponding to the initial extent of the dome can be identified. For this reason, we tentatively consider that the western part of the Yenkahe dome was situated at the present-day location of the paleo-Yasur caldera and thus at the current Yasur eruptive centre.

This indicates that the magma source is located at depth beneath the western zone and that the Yenkahe dome probably originated from this zone. This western zone of the dome has been destroyed by the formation of the paleo-Yasur caldera, which has led to the formation of a paleo-cone whose relict is still visible on the western side of the paleo-Yasur caldera. This event reveals that the western zone of the Yenkahe dome was very active and unstable, and that the magma pierced the surface of the dome at this location.

5.2. Central zone of the Yenkahe dome

To the east of the paleo-Yasur caldera, the topography of the Yenkahe is not that of an active resurgent dome. Such a dome generally exhibits a flat roof with no axial graben. In this scenario, the flat top of the dome is affected by normal faults distributed all over the roof,

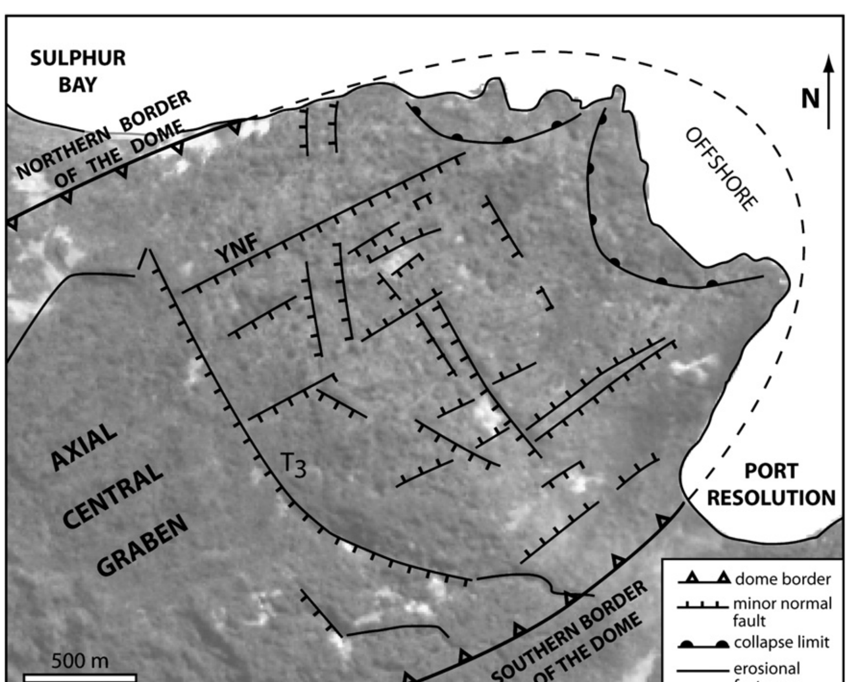

Fig. 8. The eastern zone of the dome is bounded by a large fault (offset of tens of meters, T3) indicating a relative downthrow of the western segment relative to the eastern segment. In the flat top-zone of the dome, a dense pattern of transverse and longitudinal faults is observed.

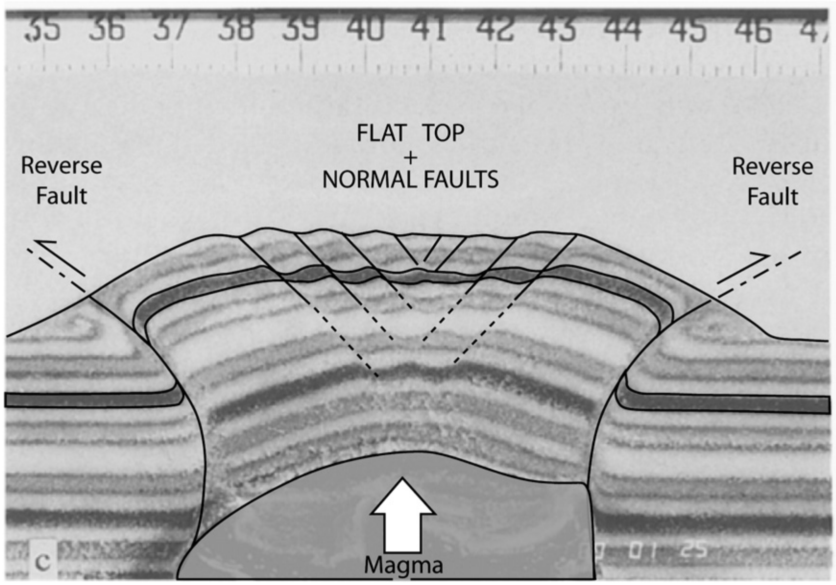

Fig. 9. Example of an analogue experiment on dome formation. The structure is limited at the base by two steeply-dipping reverse faults whereas the top of the dome is a flat zone where extension is accommodated by normal faulting (drawn from Fig. 4 in Merle and Vendeville, 1995).

which accommodate the extension generated by the uplift of the dome (Fig. 9) (Vendeville and Jackson, 1992; Schultz-Ela et al., 1993; Merle and Vendeville, 1995). As a result of this deformation, the topography of an active volcanic dome is that of an upside-down cauldron. In contrast, the topography of the Yenkahe looks like a dome whose central part (i.e. the axial graben) has collapsed between the YNF and the YSF. We conclude that the deformation in the central zone of the Yenkahe dome was two-fold: active resurgent doming followed by the collapse of its central part.

The chronology for the dome collapse and the paleo-Yasur caldera formation is difficult to establish, which we interpret as an argument that the two events are roughly coeval. This makes sense if we consider that the underlying magma may not have evolved independently on both sides of the limit between the caldera and the axial graben.

This interpretation makes it possible to reconcile the following data:

1. The ring fault is not visible between the YNF and the YSF and there is no escarpment between the flat caldera floor and the flat floor of the axial graben. This suggests that both the collapse of the caldera and that of the axial graben occurred together.
2. Normal faults in the caldera (L1, L2 and L3) converge towards the eruptive centre. According to van Wyk de Vries and Merle (1996), the local stress field due to the load of a volcano leads to the capture of normal faults that converge towards it. The changes in orientation between the YNF/YSF and the L1/L3 faults, as well as the limited offset (a few meters) of the L1/L3 faults, reveal that the YSF/YNF were already present when the L1/L3 were formed. For this reason, although L1 and L3 normal faults in the lava infill postdate the main formation of the caldera, they record the last increment of the global collapse (i.e. graben + caldera).

This indicates that the collapse process in the whole area took place over a long period of time. Thus, the collapse was a major event in the area which affected both the western and central zones of the Yenkahe dome more or less synchronously.

5.3. Eastern zone of the Yenkahe dome

The structural observations gathered in the Yenkahe dome point to the eastern zone being the only active uplift zone at the present-day. Evidence for this is: a) the flat roof, which is that of an active dome, b) the dip of marine strata along the northern flank, which reaches up to 70°, c) the highest thermal anomalies, which are concentrated there, especially along the borders, and d) the impressive uplift (>150 m), which occurred rapidly at a very recent time (i.e. the last 1000 yr).

This allows us to put forward some hypotheses about the eastern extent of the dome. It is doubtful that the initial extent of the dome to the east corresponds to its present-day termination. Recent marine deposits now on top of the dome at an elevation of about 200 m, indicate that the eastern part of the dome was still underwater a few hundred years ago. Evidence for the presence of magma beneath (high temperatures, fumaroles, altered rock, landslides, etc) show that this zone is more active than the central zone of the dome. These observations strongly suggest that the eastern part of the current dome did not exist prior to the western collapse (paleo-Yasur caldera + central graben). Recent activity (hot fumaroles and alteration) and doming in the east may be attributed to eastward migration of the underlying magma.

5.4. A tectonic model

Considering that the formation of the paleo-Yasur caldera was apparently not associated with the emission of significant aerial or effusive products, we propose that the eastward migration of the magma underneath triggered the caldera and the axial graben collapse along the ring fault and the YNF/YSF, respectively. Further eastward, escape of the underlying magma generated doming in the eastern zone. Within the axial graben, the surface deformation has been accommodated by a succession of transverse faults that allowed the uplift of the eastern zone and the coeval collapse of the central zone (Fig. 10). Despite the eastward migration of the magma, the eruptive centre (Yasur and paleo-Yasur cinder cones) is still connected with a magma chamber at depth, from which the magma now located under the eastern part of the dome is also derived.

In this scenario, an elongated resurgent dome formed prior to the occurrence of a complex collapse, which included both a circular caldera in the west (i.e. the paleo-Yasur caldera) and a linear graben in the east (i.e. the axial graben). It is not clear as to why such different mechanisms of rupture occurred at the same time. The formation of a circular caldera probably resulted from its location above the deep source of the magma, as is indicated by the continued presence of the eruptive centre in this area (paleo-Yasur and Yasur cinder cones). After the initial

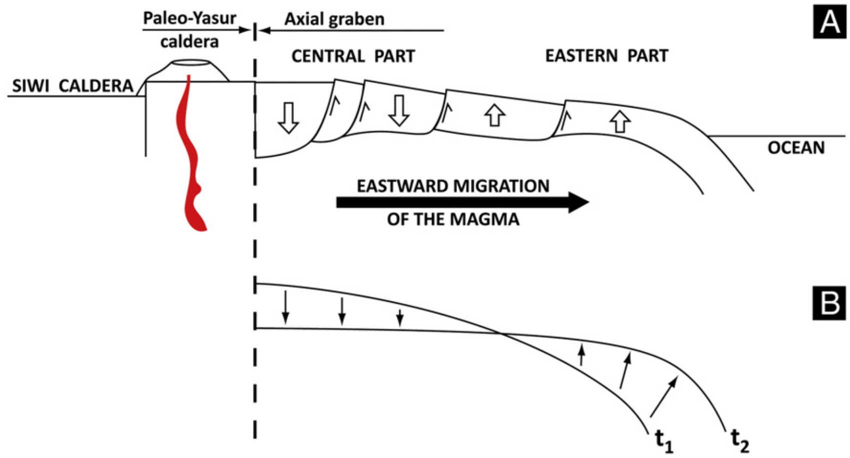

Fig. 10. A) The eastward migration of the underlying magma generates the collapse of a wide zone including the paleo-Yasur caldera and the axial central graben. B) This whole collapse to the west is coeval with the uplift of the eastern zone.

formation of an elongated resurgent dome (i.e. the Yenkahe dome), the shallow reservoir above the deep magma source may have changed its shape to become spherical, causing a circular caldera to form when the whole collapse took place.

5.5. Influence of regional tectonics

Most resurgent domes are sub-circular (Acocella et al., 2001). The structural explanation is that the ascent of the magma from a vertical cylindrical conduct generates a circular intrusion when approaching the Earth's surface. On the other hand, the formation of an elongated dome must be related to an elongated reservoir underneath. This raises the question as to how a vertical magma conduit could create an elongated reservoir near the surface.

It is tempting to relate the elongated shape of the reservoir to the regional tectonics resulting from the subduction of the Indo-Australian plate. We note that the Yenkahe dome is strongly elongated in the direction of the regional stretching, i.e. in the direction normal to the main trend of the subduction zone located to the west of the island, and normal to the back-arc basin of Futuna to the east. This cannot be purely coincidental.

The growth of an elongated reservoir could be related to a lateral unconfined boundary. This unconfined boundary could be due both to the steep eastward offshore topographic slope towards the deep North-Fiji back-arc basin and to the (regional) weak lateral confining stress in this direction. The magma rises vertically up to the point where the unconfined boundary causes it move horizontally in this direction. This creates an elongated dome at the surface. During a second phase, further lateral escape of the magma triggers collapse of this elongated structure and doming to the east, in the same unconfined direction.

The Siwi caldera also shows a clear elongation in the same direction suggesting that the whole volcano-tectonic history in this area is controlled by the regional tectonics related to the subduction zone.

6. Concluding remarks

The main conclusions of this structural study can be summarized as follows:

1. A resurgent dome in the heart of a large caldera has undergone a long and complex volcano-tectonic history, associated with eruptions, caldera formation and lateral migration of the underlying magma.
2. The elongated resurgent dome may have developed due to the ascending magma intruding laterally near to the surface as a result of a lateral unconfined boundary.
3. Further lateral escape of the magma may have triggered the collapse of the elongated resurgent dome.
4. The collapse of the elongated dome was complex and may have generated a sub-circular caldera above the deep source of the magma and a graben in the adjacent part of the dome.
5. In the Siwi caldera of Tanna Island, magma migration and the formation of an active dome at the coast present a real risk for inhabitants of nearby villages (Ipikel and Port Resolution). We consider this risk to be higher than that posed by to the current Yasur volcano activity.

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